GATE ME 90 Days Challenge

Day 51/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Engineering Mathematics	**Complex Variables & Numerical Methods**	**52 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on analytic functions and the Cauchy–Riemann equations, harmonic conjugates, singularities and residues, Cauchy's integral theorem and formula, Laurent series; Newton–Raphson and bisection, the trapezoidal and Simpson's rules, and Euler, Heun and Runge–Kutta methods for ODEs. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

With $z = x + iy$, which of the following functions is NOT analytic?

(A) $\sin z$

(B) $z^3 + 2z$

(C) e^{2z}

(D) $x^2 + iy^2$

Q2 NAT

$f(z) = u + iv$ is analytic with $u = x^3 - 3xy^2$ and $v(0, 0) = 0$. The value of v at the point (1, 2) is _____

Q3 MCQ

The residue of $f(z) = e^{2z}/(z - 1)^2$ at $z = 1$ is

(A) $2e^2$

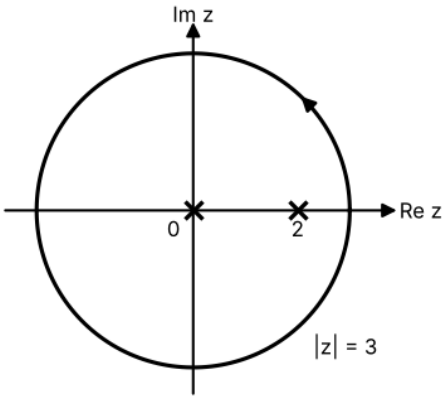
(B) e^2

(C) $e^2/2$

(D) 0

Q4 NAT

Let C be the circle $|z| = 3$, traversed anticlockwise, and $I = \oint_C (z^2 + 1)/[z(z - 2)]\, dz$. The value of $I/(\pi i)$ is _____



Q5 MCQ

The function $f(z) = e^{1/z}$ has at $z = 0$

(A) a simple pole
(B) a removable singularity
(C) an essential singularity
(D) a pole of order 2

Q6 NAT

The cube root of 20 is found by the Newton–Raphson method applied to $f(x) = x^3 - 20$, starting from $x_0 = 3$. The value of x_1 (round off to four decimal places) is _____

Q7 MCQ

Simpson's 1/3 rule can be applied directly (without combining with another rule) only when the interval is divided into

(A) an odd number of strips
(B) an even number of strips
(C) a multiple of three strips
(D) any number of strips

Q8 NAT

The integral of sin x from 0 to π is evaluated by the trapezoidal rule with four equal strips ($h = \pi/4$). The value obtained (round off to three decimal places) is _____

Q9 MCQ

For Euler's explicit method with step size h, the local truncation error and the global error are of order

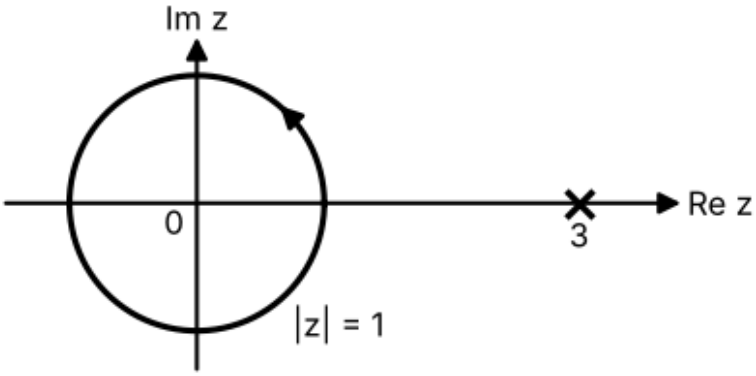
(A) h^2 and h respectively
(B) h and h^2 respectively
(C) h^3 and h^2 respectively
(D) h^2 and h^2 respectively

Q10 NAT

The classical fourth-order Runge–Kutta method is used for $dy/dx = -2xy$, $y(0) = 1$, with one step $h = 0.2$. The value of y at $x = 0.2$ (round off to four decimal places) is _____

Q11 MCQ

Let C be the circle $|z| = 1$ traversed anticlockwise, as shown. The value of $\oint_C z^2/(z-3)\,dz$ is



(A) $2\pi i \times 9$
(B) $2\pi i \times 3$
(C) πi
(D) 0

Q12 NAT

The integral of e^{-x^2} from 0 to 1 is evaluated by Simpson's 1/3 rule with four equal strips (h = 0.25). The value obtained (round off to four decimal places) is _____

Q13 MCQ

The function $f(z) = (\sin z)/z^4$ has at $z = 0$

(A) a pole of order 4
(B) a removable singularity
(C) a pole of order 3
(D) an essential singularity

Q14 NAT

The bisection method is applied to $f(x) = x^3 - x - 1$ on the interval [1, 2]. The third midpoint computed (m_3, round off to three decimal places) is _____

Q15 MCQ

In polar form, $z = re^{i\theta}$ and $f = u(r, \theta) + iv(r, \theta)$, the Cauchy–Riemann equations are

(A) $u_r = v_\theta$ and $u_\theta = -v_r$
(B) $u_r = (1/r)v_\theta$ and $v_r = -(1/r)u_\theta$
(C) $u_r = r\,v_\theta$ and $v_r = -r\,u_\theta$
(D) $u_r = -(1/r)v_\theta$ and $v_r = (1/r)u_\theta$

Answer Key

Q	Type	Answer
Q1	MCQ	(D) $x^2 + iy^2$
Q2	NAT	-2
Q3	MCQ	(A) $2e^2$
Q4	NAT	4
Q5	MCQ	(C) an essential singularity
Q6	NAT	2.7407 (accept 2.7380 to 2.7435)
Q7	MCQ	(B) an even number of strips
Q8	NAT	1.896 (accept 1.890 to 1.902)
Q9	MCQ	(A) h^2 and h respectively
Q10	NAT	0.9608 (accept 0.9600 to 0.9616)
Q11	MCQ	(D) 0
Q12	NAT	0.7469 (accept 0.7460 to 0.7477)
Q13	MCQ	(C) a pole of order 3
Q14	NAT	1.375 (accept 1.374 to 1.376)
Q15	MCQ	(B) $u_r = (1/r)v_\theta$ and $v_r = -(1/r)u_\theta$

Detailed Solutions

Q1 **Answer: (D) $x^2 + iy^2$**

Concept: $f = u + iv$ is analytic in a region only if $u_x = v_y$ and $u_y = -v_x$ hold there.

For $x^2 + iy^2$: $u_x = 2x$, $v_y = 2y$, so C–R hold only on the line $y = x$.

A function differentiable only on a line has no neighbourhood of analyticity: **(D)**.

Exam tip: Any function built from z alone (polynomials, e^z, sin z) is analytic; anything that needs x and y separately is suspect.

Q2 **Answer: -2**

Concept: Find the harmonic conjugate from the C–R equations.

$v_y = u_x = 3x^2 - 3y^2 \rightarrow v = 3x^2y - y^3 + g(x)$

$v_x = 6xy + g'(x)$ must equal $-u_y = 6xy$, so $g = \text{const} = 0$

$v(1, 2) = 3(1)(2) - 8 = \mathbf{-2}$ (indeed $f = z^3$)

Exam tip: Milne–Thomson shortcut: $f'(z) = u_x(z, 0) - i\, u_y(z, 0) = 3z^2$, so $f = z^3$.

Q3 **Answer: (A) $2e^2$**

Concept: For a pole of order 2 at a: Res = $d/dz[(z - a)^2 f(z)]$ at $z = a$.

$(z - 1)^2 f = e^{2z}$; its derivative is $2e^{2z}$

At $z = 1$: Res = **$2e^2$ (A)**

Exam tip: For order m: Res = $[1/(m - 1)!]\, d^{m-1}/dz^{m-1}[(z - a)^m f]$ at a.

Q4 **Answer: 4**

Concept: Residue theorem: $I = 2\pi i \times$ (sum of residues inside C).

Both $z = 0$ and $z = 2$ lie inside $|z| = 3$.

Res(0) $= (0 + 1)/(0 - 2) = -1/2$; Res(2) $= (4 + 1)/2 = 5/2$; sum $= 2$

$I = 2\pi i \times 2 = 4\pi i$, so $I/(\pi i) =$ **4**

Exam tip: Always check which singularities lie inside the contour before adding residues.

Q5 **Answer: (C) an essential singularity**

Concept: Classify by the principal part (negative powers) of the Laurent series.

$e^{1/z} = 1 + 1/z + 1/(2!z^2) + 1/(3!z^3) + \ldots$ has infinitely many negative powers.

So $z = 0$ is an **essential singularity (C)**; its residue is 1.

Exam tip: Finite number of negative powers → pole (order = highest power); none → removable.

Q6 **Answer: 2.7407 (accept 2.7380 to 2.7435)**

Concept: $x_{n+1} = x_n - f(x_n)/f'(x_n)$.

$f(3) = 27 - 20 = 7$; $f'(3) = 3 \times 9 = 27$

$x_1 = 3 - 7/27 =$ **2.7407**

Exam tip: The true root is 2.7144; one more step gives 2.7146, showing quadratic convergence.

Q7 **Answer: (B) an even number of strips**

Concept: Simpson's 1/3 rule fits a parabola through each pair of strips (three points).

So the number of strips n must be **even (B)**.

Simpson's 3/8 rule uses cubics over groups of three strips (n a multiple of 3).

Exam tip: $I \approx (h/3)[f_0 + 4$(odd ordinates) $+ 2$(even interior ordinates) $+ f_n]$.

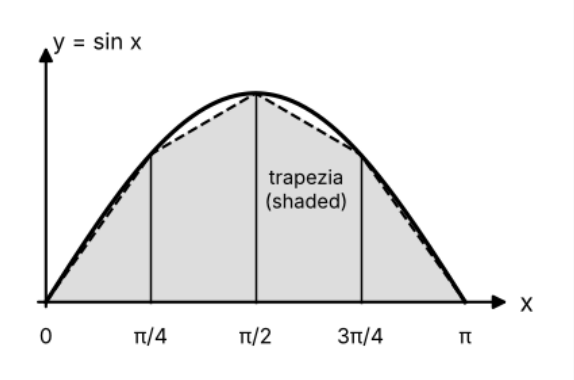
Q8 **Answer: 1.896 (accept 1.890 to 1.902)**

Concept: $I \approx (h/2)[f_0 + 2(f_1 + f_2 + f_3) + f_4]$.

Ordinates: 0, 0.7071, 1, 0.7071, 0

$I \approx (\pi/8)[0 + 2 \times 2.4142 + 0] = 0.3927 \times 4.8284 =$ **1.896**

Exam tip: The exact value is 2; for a concave function the trapezoidal rule underestimates (chords lie below the curve).



Q9 **Answer: (A) h^2 and h respectively**

Concept: Euler's method keeps only the first two Taylor terms: $y(x + h) = y + hy' + (h^2/2)y'' + \ldots$

Error per step $\propto h^2$; over $(b - a)/h$ steps the errors add up to $\propto h$.

So **local $O(h^2)$, global $O(h)$ (A)**: Euler is a first-order method.

Exam tip: Heun/modified Euler: global $O(h^2)$; classical RK4: global $O(h^4)$.

Q10 **Answer: 0.9608 (accept 0.9600 to 0.9616)**

Concept: $y_1 = y_0 + (h/6)(k_1 + 2k_2 + 2k_3 + k_4)$.

$k_1 = f(0, 1) = 0$; $k_2 = f(0.1, 1) = -0.2$

$k_3 = f(0.1, 0.98) = -0.196$; $k_4 = f(0.2, 0.9608) = -0.38432$

$y_1 = 1 + (0.2/6)(0 - 0.4 - 0.392 - 0.38432) = 1 - 0.039211 =$ **0.9608**

Exam tip: The exact solution e^{-x^2} gives 0.960789 at $x = 0.2$: RK4 agrees to six decimals.

Q11 **Answer: (D) 0**

Concept: Cauchy–Goursat theorem: the integral of a function analytic inside and on a closed contour is zero.

The only singularity, $z = 3$, lies outside $|z| = 1$.

So the integrand is analytic on and inside C and the integral is **0 (D)**.

Exam tip: $2\pi i \times 9$ would be the answer only if the contour enclosed $z = 3$ (e.g. $|z| = 4$).

Q12 **Answer: 0.7469 (accept 0.7460 to 0.7477)**

Concept: $I \approx (h/3)[f_0 + 4(f_1 + f_3) + 2f_2 + f_4]$.

$f_0 = 1$, $f_1 = 0.939413$, $f_2 = 0.778801$, $f_3 = 0.569783$, $f_4 = 0.367879$

Sum $= 1 + 4(1.509196) + 2(0.778801) + 0.367879 = 8.962266$

$I \approx (0.25/3) \times 8.962266 =$ **0.7469**

Exam tip: The exact value is 0.746824: Simpson's rule with only four strips is already accurate to four decimals.

Q13 **Answer: (C) a pole of order 3**

Concept: Expand and read off the highest negative power.

$\sin z/z^4 = (z - z^3/6 + z^5/120 - \ldots)/z^4 = 1/z^3 - 1/(6z) + z/120 - \ldots$

The highest negative power is z^{-3}: a **pole of order 3 (C)**, with residue $-1/6$.

Exam tip: Order of pole = (order of zero of denominator) − (order of zero of numerator) $= 4 - 1$.

Q14 **Answer: 1.375 (accept 1.374 to 1.376)**

Concept: Keep the half-interval at whose ends f changes sign.

$m_1 = 1.5$: $f = 0.875 > 0$, $f(1) = -1 < 0 \rightarrow$ root in $[1, 1.5]$

$m_2 = 1.25$: $f = -0.297 < 0 \rightarrow$ root in $[1.25, 1.5]$

$m_3 =$ **1.375** ($f = 0.2246 > 0$, so the next interval is $[1.25, 1.375]$)

Exam tip: After n bisections the error bound is $(b - a)/2^n$; the true root is 1.3247.

Q15 **Answer: (B) $u_r = (1/r)v_\theta$ and $v_r = -(1/r)u_\theta$**

Concept: Equate f' computed along the radial and the circumferential directions.

Radial: $f' = e^{-i\theta}(u_r + iv_r)$; circumferential: $f' = e^{-i\theta}(v_\theta - iu_\theta)/r$

Hence **$u_r = v_\theta/r$ and $v_r = -u_\theta/r$ (B)**.

Exam tip: Check with $f = z^2 = r^2e^{2i\theta}$: $u_r = 2r\cos 2\theta = (1/r)(2r^2\cos 2\theta) = v_\theta/r$.

What next?

Day 51 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 52/90: Applied Thermodynamics, Turbomachinery. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.